

In electronics, some of the most sought-after training technologies, as mentioned by industry experts, include IoT kits, PC interface products in digital and analogue electronics, all-in-one electronic test benches and so on. In electrical engineering, latest products include PC interface electrical machine trainers and PC interface PID trainers.

Talking about the most in-demand training instruments, Dr Pravin Raut, founder and managing director, SINCOM Sindhu Electronics and Communications Pvt Ltd, says, "The Indian educational system and market demand both traditional training products as well as modern trainer kits. Traditional systems are based on analogue electronics to wireless communication.

"On the other hand, some modern trainer tools are digital communication and radio frequency (RF) wireless communication kits.

"In addition, looking at our inventory, we see a high demand of audio and video labs, digital communications and electronic components, display boards, digital voltage and current (DVI) meters and digital multi voltage and current (DMVI) meters."

Do-it-yourself (DIY) boards are in great demand as well. Modular IoT development boards and sensors are seeing great popularity. Many kits in demand today fall under DIY tools category. Some popular kits are Arduino, Raspberry Pi, Blockly and Beagle Bone, among others.

Parimal Wagh, senior administration officer, Center for Development of Advanced Computing (CDAC), Pune, says, "Training programmes can make use of open source tools and open hardware to provide necessary guidance. Raspberry Pi3, Beagle Bone Black, Thunderboard React, Freedom



Raspberry Pi board

Board and Zedboard are some boards we use."

Many modern training kits, especially for embedded or automated electronics projects, are based on Arduino, mainly due to its cost-effectiveness and simple learning curve. Moreover, there is a lot of learning material available for Arduino, using which one can easily scale up the skill level. Jayesh Jain, director, Macfos Pvt Ltd, says that, quite a few of these kits are available for as low as ₹ 1000.

Wi-Fi modules, 3D printers and drones are some other sought-after products for hands-on training purposes.

Augmented reality (AR)- and virtual reality (VR)-based training platforms are other major trends. Both generate a digitally simulated environment that allows users to study subjects or even practice hands-on applications digitally, using a VR headset and a software platform that simulates the training environment. These training models bring new users up to speed with actual prototype building, without using physical resources and, eventually, making them production-ready.

Shaping newcomers with newer amenities

Over the years, there has been concern among industry experts about India's educational setups with regards to whether the institutions are providing relevant courses and training to make the students industry-ready.

Ashwin Agarwal, president - strategy and operations, Phi Education Solutions, shares, "Currently, educational programmes are neither intensive, hands-on nor linked to industry needs. The market is new and evol-

ing. Hence, training programmes need to catch up."

However, educational setups are steadily aligning more with industry standards, aiming to shape the students into production-ready engineers.

Dr Raut explains, "These days, most universities have revised their syllabi according to the need of modern technological trends, with the aim to make newcomers more employable. The curricula now include advanced embedded systems, 4G mobile communications, WSN, robotics, mechatronics, interfacing of IoT with electronic systems and so on. Also, various software EDA tools that are widely used for design, simulation and interfacing of circuits are being taught."

For instance, C-DAC Pune provides post graduation diploma courses on subjects like embedded system design (PG-DESD), the Internet of Things (PG-DIoT), big data analytics (PG-DBDA) and others.

Since latest technologies employ both software and hardware skills proportionately, it is crucial for today's training tools to provide an access point to both.

Jain says, "Logic development for programming can be started with kids as young as 10 years of age. Many kits are now based on scratch programming for Arduino, which allow kids to create flowchart diagrams on tablets or PCs. The platform automatically creates and programs hardware based on those flowcharts. These new advancements in kits allow users to directly start working on their projects or new, emerging technologies without wasting much time in learning low-level programming and electronics."

As modern technologies, like the IoT, embedded electronics, automation or AI, become more prevalent, students and newcomers need to be given the right practical knowledge and hands-on training that can help them jump into the industry and contribute without further re-skilling. This benefits them in their career and the industry in their business, alike. Essential investments allotted to training and supervising from scratch can be saved.

Modern training kits are setting these initiatives in motion as these become more relevant to the smart



Arduino-based kits are gaining high popularity